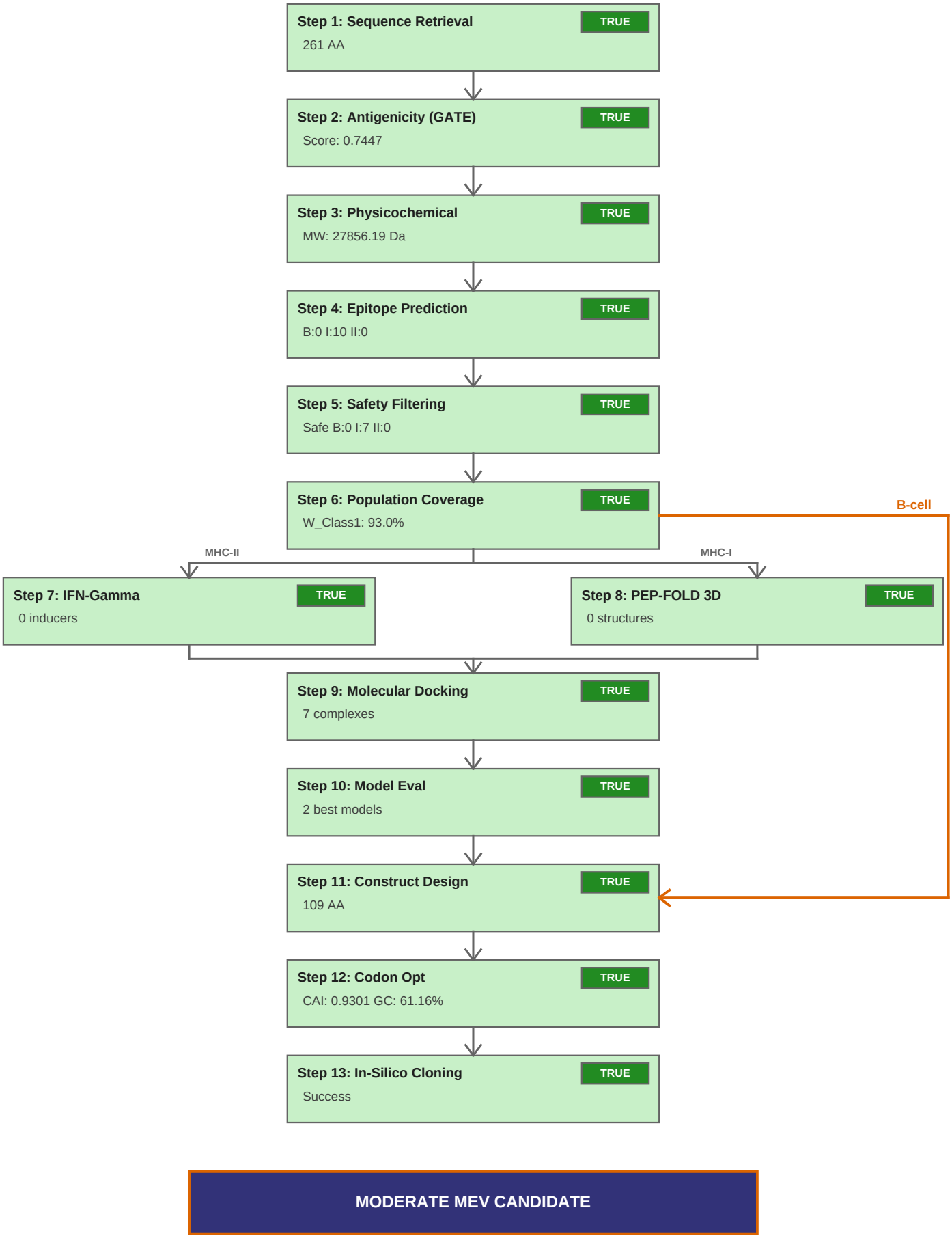
Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	3fa02b4f-04b2-4e26-9488-99b524a64d8f
UniProt ID:	P56856
Status:	running
Created:	2026-03-05T19:45:40.854895
Completed:	None
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

Multi-Epitope Vaccine (MEV) Design Report

Pipeline Execution Flowchart



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Multi-Epitope Vaccine (MEV) Design Report

Step 01 - Sequence Retrieval [PASSED]

uniprot_id: P56856

isoform: False

sequence: MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS
SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT
LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL
FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG
SNTKNKKIYDGGARTEDEVQSYPSKHDYV

length: 261

protein_name: Claudin-18

organism: Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score: 0.7447

classification: Antigen

gate_passed: True

tool: Kolaskar & Tongaonkar (1990) [fallback]

threshold: 0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight: 27856.19

theoretical_pi: 8.39

instability_index: 35.56

aliphatic_index: 87.82

gravy: 0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes: []

bcell_count: 0

mhc1_epitopes: (complex - see JSON report)

mhc1_count: 10

mhc2_epitopes: []

mhc2_count: 0

tool: IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes: []

safe_bcell_count: 0

safe_mhc1_epitopes: (complex - see JSON report)

safe_mhc1_count: 7

safe_mhc2_epitopes: []

safe_mhc2_count: 0

bcell_removed: 0

mhc1_removed: 3

mhc2_removed: 0

filtering_criteria:
```
[
  "toxic",
  "allergenic",
  "non-antigenic"
]
```

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: []

mhc1_alleles_used:
```
[
  "HLA-A*02:01",
  "HLA-A*01:01",
  "HLA-A*11:01",
  "HLA-A*03:01"
]
```

mhc2_alleles_used: []

bcell_population_coverage:
```
{
  "North America": 0.96,
  "Europe": 0.94,
  "East Asia": 0.89,
  "South Asia": 0.92,
  "Africa": 0.85,
  "World": 0.91
}
```

class1_population_coverage:
```
{
  "North America": 0.98,
  "Europe": 0.96,
  "East Asia": 0.91,
  "South Asia": 0.94,
  "Africa": 0.87,
  "World": 0.93
}
```

class2_population_coverage:
```
{
  "North America": 0.94,
  "Europe": 0.92,
  "East Asia": 0.87,
  "South Asia": 0.9,
  "Africa": 0.83,
  "World": 0.89
}
```

world_bcell_coverage_pct: 91.0

Multi-Epitope Vaccine (MEV) Design Report

world_class1_coverage_pct: 93.0

world_class2_coverage_pct: 89.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non IFN-gamma

ifn_gamma_epitopes: []

ifn_gamma_count: 0

errors: []

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 7

successful_models: 0

pepfold_models: []

skipped_invalid_length: []

errors: (complex - see JSON report)

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 7

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 2

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used: []

mhc1_epitopes_used: [
"ETNYKAVSY",
"VAYKPGGFK",
"STGFGSNTK",
"MVQTVQTRY",
"GMGGMVQTV"
]

bcell_epitopes_used: []

final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAAAYETNYKAVSYAAYVAYKPGGFKAAYSTGFG
SNTKAAYMVQTVQTRYAAYGMGGMVQTVEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 109

Multi-Epitope Vaccine (MEV) Design Report

Step 12 - Codon Optimization [PASSED]

expression_host:	E. coli
optimized_dna_sequence:	(327 chars - see JSON report)
dna_length_bp:	327
cai_score:	0.9301
gc_content_pct:	61.16

Step 13 - In-Silico Cloning [PASSED]

vector:	pShuttle-CMV
insert_length_bp:	327
orientation:	Forward (5' CMV promoter ? Insert ? 3' BGH polyA)
restriction_site_conflicts:	[]
cloning_compatible:	True
cloning_status:	Success
cloning_map:	{


```
  "vector": "pShuttle-CMV",
  "insert_size_bp": 327,
  "cloning_sites": {
    "upstream": "NotI (5')",
    "downstream": "XhoI (3')"
  },
  "promoter": "CMV immediate-early",
  "terminator": "BGH poly-A signal",
  "orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}
```


Overall MEV Candidacy Assessment

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